

William Claytor

Senior Quality Engineer and Automation Leader

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Summary

Senior Quality Engineer and automation leader with 25+ years of experience transforming manual QA into scalable automated systems, currently specializing in AI-assisted QA workflows and testing generative AI applications. Proven track record of building and modernizing test frameworks (Playwright, Selenium), leading globally distributed QA teams, and embedding quality into CI/CD pipelines across SaaS, enterprise, public-sector, and regulated environments.

Skills

Quality Engineering: Test automation architecture, test strategy, generative AI testing, AI-assisted QA workflows, API testing, performance testing, cross-platform testing, requirements traceability, QA team leadership

Test Automation Tools: Playwright, Selenium, Nightwatch.js, Appium, JMeter, Gatling, RSpec

Quality Management Tools: Jira, Xray, Testiny, TestRail

Languages: TypeScript, JavaScript, Python, Ruby, C#, Java, PowerShell, SQL

CI/CD & Infrastructure: AWS, Azure, GitHub Actions, Jenkins, Docker

Experience

Cloudforce, Cloud Applications Test Engineer — National Harbor, MD
Feb 2026 – Oct 2026
9 months

- Designed and executed detailed test plans and test cases for nebulaONE, an Azure-hosted generative AI platform for public-sector and higher-education organizations
- Modernized the Playwright/TypeScript end-to-end test automation suite with page objects, shared helpers, consistent locators, state-based assertions, and context-based tags for targeted execution
- Automated tests for AI agents using retrieval-augmented generation (RAG), API and Model Context Protocol (MCP) connections, web search, and multi-agent collaboration
- Validated LLM responses for knowledge-grounded agents, asserting source usage, citations, and expected content
- Built a requirements traceability matrix linking feature areas and sub-features to Jira tickets, Testiny cases, and Playwright tests to establish measurable feature coverage
- Implemented AI-assisted QA workflows with Claude Code and GitHub Copilot for requirements analysis, test generation, and pull request review, supported by PowerShell scripts integrating Jira, Testiny, and GitHub APIs
- Trained and mentored QA and engineering team members in Playwright and AI-assisted test automation

insightsoftware, Senior QA Automation Engineer and QA Lead — Remote
May 2023 – Dec 2025
2 years 8 months

- Executed sprint release verification testing across multiple product lines using Azure Pipelines with an in-house C#/Selenium automation framework
- Built a daily Azure test-results reporting tool that reduced report preparation from 10+ minutes to under 30 seconds
- Built a pipeline migrating 5,000+ automated test cases from Team Foundation Server into Jira/Xray
- Implemented an AI-assisted QA workflow using GitHub Copilot and PowerShell to automate Jira/Xray operations across requirements analysis, test execution, and reporting
- Served as a QA Lead for distributed QA teams, establishing onshore/offshore processes and guiding manual testers into automation
- Coordinated customer-facing disaster recovery testing and release test reporting across distributed teams
- Established performance testing capability by leading the initial development of a JMeter/Grafana framework and helping hire and onboard Principal and Senior Performance Engineers

Independent Work, Professional Development Sabbatical — July 2022 – May 2023
11 months

Built and deployed Songbook, a Rails setlist organizer, using Dokku and DigitalOcean; evaluated Playwright and Cypress through Page Object Model implementations.

Active911, Senior QA Engineer and QA Manager — Corvallis, OR
Mar 2020 – July 2022
2 years 5 months

- Established end-to-end QA coverage for first-responder alerting and scheduling applications across web, Android, and iOS
- Architected a Nightwatch.js framework with page objects and reusable modules to accelerate test development
- Built a real-time test automation dashboard integrating Jenkins Job DSL with the TestRail API for live execution status and reporting
- Grew and mentored the QA team through hiring, internal promotions from Customer Support, and automation training
- Embedded quality in development processes by collaborating with product and engineering during backlog review and design sessions
- Coordinated UAT with product managers, ensuring customer requirements were validated before releases

Puppet, Senior Software Engineer — Portland, OR
Jan 2018 – Mar 2020
2 years 3 months

- Extended the Gatling-based Puppet load-test framework with scale testing, HTML comparison reports, and system tuning integration, backed by RSpec unit tests
- Built a Ruby-based provisioning utility to automate infrastructure node deployment through the Puppet Automated Build System API
- Contributed to the Reference Architecture Setup CLI using Ruby and Puppet Bolt, implementing Docker-based acceptance tests with RSpec coverage
- Drove performance testing standards through the Reference Architectures Working Group, collaborating with the Customer Success, Installer, and PuppetDB teams on the architecture definition and test roadmap
- Served as Scrum Master for the Installer and System Level Verification teams, establishing the Puppet Operating Model implementation and facilitating continuous process improvement

National Public Radio, Senior Quality Engineer — Washington, DC
May 2014 – Jan 2018
3 years 9 months

- Led a cross-team JavaScript framework evaluation and implemented Nightwatch.js cross-browser testing in the CMS CI pipeline
- Architected a keyword-based data-driven Java/Selenium framework for cross-platform testing
- Developed automated tests for Android and iOS using Appium with a Jenkins-based simulator and emulator infrastructure
- Established Jenkins automation dashboards using Job DSL across the Publishing, TechOps, and NPR One teams
- Modernized the API test suite with modular architecture, adding coverage for new end-points and improving maintainability

American Public University System, QA Automation Manager — Charles Town, WV
Mar 2012 – May 2014
2 years 3 months

- Designed a full-stack test automation architecture in .NET / C# including a keyword-driven UI framework (Coded UI + Selenium) plus data-driven back-end framework verifying student registration and payment transactions
- Built the QA Automation team through recruiting for engineer and specialist roles and training team members in automation tools, development practices, and organizational processes
- Established an Agile test management workflow using HP ALM / Quality Center for work-load planning and execution tracking
- Drove automation adoption organization-wide through stakeholder engagement and active participation in design, requirements, and test plan reviews

FINRA, Lead QA Analyst — Rockville, MD
Sept 2003 – Mar 2012
8 years 7 months

- Ensured the quality of Web CRD (the central registration and licensing system for the U.S. securities industry) and IAPD (Investment Adviser Public Disclosure)
- Evolved the test automation platform from WinRunner to QTP, architecting improved dependency management for multi-application, multi-release, multi-environment testing
- Designed a no-code test framework (QTP / VBScript + Excel parameterization) enabling manual testers to create and execute automated tests without programming knowledge
- Pioneered CI adoption, implementing automated test execution pipelines using CruiseControl and later Hudson (now Jenkins) with Ant scripting
- Established a requirements-based automation process with HP Quality Center for traceability, test organization, and defect management

Earlier Experience

Test Automation Specialist, Fannie Mae — Washington, DC | 2002 – 2003

Test Automation Specialist, MedContrax — Gaithersburg, MD | 2001 – 2002

Test Automation Specialist, US Census Bureau — Suitland, MD | 1998 – 2001

Education

Towson University, English Studies — Towson, MD
1992 – 1994

Harford Community College, Business Management — Churchville, MD
1990 – 1991